

ECON 164: Theory of Economic Growth

Week 9B: Review

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2. **review our common toolkit**
 - Cobb-Douglas production function w/ labor-augmenting technology
 - factor accumulation: $\dot{K}$, $\dot{L}$, $\dot{A}$
 - steady states, phase diagrams, BGPs, ...

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 - steady states, phase diagrams, BGPs, ...
3. **practice!** esp. comparative statics

Three organizing questions

1. Why are we so rich and they so poor?
2. What is the engine of economic growth?
3. How do “growth miracles” happen?

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→ differences in s_I ($\beta, \tau, \dots$), g_L , A_0 (or $D_0, B_0, \dots$)

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3. How do “growth miracles” happen?

→ transitional growth from **physical capital** as in the NGM...

... and transitional growth from **technology** through R&D, adoption, and trade

Our approach: The cosmologists of economics

- one universe, one global economy → **no controlled experiments**
- instead a **back-and-forth between data and models**

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- A **model** is a **mathematical representation** of some aspect of the economy...
 - agents' objectives, constraints
 - market structure, feasibility conditions
 - parameters, exogenous variables
- ... that determines how we **interpret data** and **evaluate policy**

Two building blocks of every model

1. *How do economies produce?*
2. *How do economies accumulate factors of production?*

The production function

1. neoclassical growth model (Solow + Ramsey-Cass-Koopmans)

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5. **natural resources model** (w/ energy and land)

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- Hicks-neutral vs. Harrod-neutral
- Malthusian model

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... but the nuts-and-bolts are the same!

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The stage we've set

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Growth can either be from...

- getting more K
- getting more L
- improving A

Accumulating factors

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$$\dot{R}_t = -s_E R_t$$

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$$\dot{K}_t = s_I Y_t - \delta K_t$$

$$\dot{L}_t = g_L L_t$$

$$\dot{A}_t = \dots$$

...depends on the particular model!

What does it mean to “solve” the model?

Given. . .

- initial levels $\{K_0, A_0, L_0\}$
- parameters $\{\alpha, \delta, g_L, \dots\}$
- equations of the model

in class: with phase diagrams!

. . . we obtain a **time path** for the **endogenous** variables $\{K_t, Y_t, \dots\}$ expressed in terms of the **exogenous** variables $\{L_t, \dots\}$, parameters, and initial conditions.

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Growth rates: “take logs and derivatives”

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3. look at the right-hand side: is there a **ratio** of exogenous vs. endogenous variables?

$$g_K = s_I \left(\frac{K_t}{A_t L_t} \right)^{\alpha-1} - \delta, \quad g_A = \theta s_R^\lambda \left(\frac{L_t^\lambda}{A_t^{1-\phi}} \right)$$

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4. in steady state, g_Z must be **constant** $\rightarrow$ growth rate of that ratio must be **zero**, so take logs and derivatives of the ratio and set equal to zero:

$$\frac{K_t}{A_t L_t} \rightarrow g_K - (g_A + g_L) = 0 \quad \frac{L_t^\lambda}{A_t^{1-\phi}} \rightarrow \lambda g_L - (1 - \phi) g_A = 0$$

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5. **draw your phase diagram**: ratio on the x -axis, g_Z on the y -axis

Finding the BGP

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8. recall what we derived re: **GDP per capita** (may need to include h , **s_R** , ...)

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10. ...and, along the BGP,

$$y_t^{\text{BGP}} = \left(\tilde{k}^{ss}\right)^\alpha A_t^{\text{BGP}}$$
$$\ln y_t^{\text{BGP}} = \alpha \ln \tilde{k}^{ss} + \ln A_0 + g_A^{ss} t$$

References

Weinberg, Steven. 2018. *Third thoughts*. Cambridge, MA: Harvard University Press.